ECDLP lead atlas: review of 2026-10-30

Question. What changed in the three checked-out repositories between the 2026-10-29 review (`docs/reports/ecdlp-lead-atlas-20261029/`) and this session's checkouts? Which leads do those changes advance?

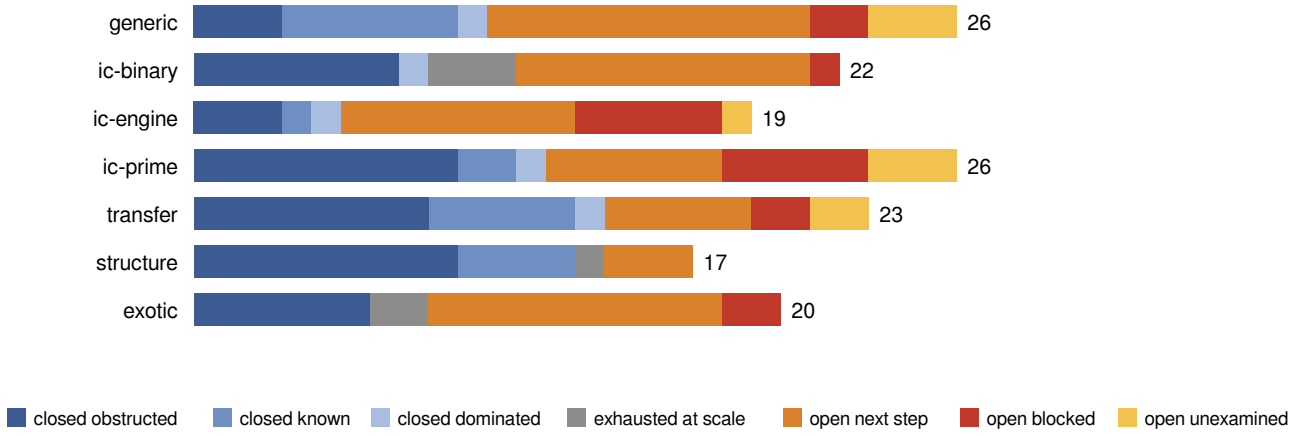
Scope and method. One session ran `python3 tools/lead_atlas.py --delta` against the 2026-10-29 pins and folded every listed path into its owning lead or recorded it as not about ECDLP. That was 469 watched paths: 341 in crypto-autoresearcher, 48 in crypto and 80 in cryptanalysis. The review covers each repository up to the commit checked out in this session:

repository	reviewed commit
crypto-autoresearcher	`ed4ded06` (main)
crypto	`3fe55f62`
cryptanalysis	`eefacd27`

Later main commits of crypto and cryptanalysis were not checked out. The container's disk allowance was spent: the crypto update alone touches 15,607 files. Those commits appear in the next `--delta`. ecc-challenges was not checked out, and its pin is carried over from 2026-10-24.

Status of each statement. Dispositions are derived from committed evidence and cite it. They are not new measurements, and they change no hypothesis, experiment or goal status. Numbers are measured values copied from the cited records.

ECDLP lead atlas, 2026-10-30: 153 leads by family and disposition



Result

- **153 leads** (149 at the 2026-10-29 stamp, 152 before this review).
  - 74 closed with a named obstruction: 45 obstructed, 18 known, 5 dominated, 6 exhausted at scale.
  - 53 open with a decided next step, 17 blocked, 9 unexamined.
- **Coverage 2100/2100.** This review added 14 rows to `coverage.tsv`.
  - Two hypotheses were simply unmapped (H-ECDLP-8dfef7, H-ECDLP-98a95a); each names its lead in `prior_art`.
  - Twelve ECDLP records sat outside the tool's area filter. The hypotheses H-CMFB, H-PFIVE, H-PNEXT, H-PSEARCH and H-SEMAEV use area prefixes missing from `ecc_areas`. The ideas IDEA-20261029-* and IDEA-20261030-* are one-line JSON, which the area regex cannot read. They are mapped by hand, and the filter gap remains in `tools/lead_atlas.py`.
- **One new lead.** `LEAD-EXO-hashed-factor-base-tensor-ecdl` (open_next_step) covers the native implementation of Akshima et al.'s hashed-factor-base ECDL (Bochum v0.2) in crypto and cryptanalysis.
  - Its subexponential bound is conditional on the unverified openai/math #k-SAT preprint (KN-LIT-caa51d).

  - Only small-case correctness is checked. Matched measurement is blocked.
  - The next step is an independent audit. It must test whether the obstruction recorded in `LEAD-ICB-decomposition-encoding-and-sat` applies: there the #SAT exponent loses to enumeration at Semaev-scale variable counts.
- **No disposition change** among existing leads. Evidence folded:
  - `LEAD-ICP-coincidence-harvest-structured-base`: the EXP-PFDR-bb0d26 review is complete. H-PFDR-727bfc is supported (EV-PFDR-07d608, strong): log-GAP x-labels match the uniform null in 44 of 45 cells (KN-FIND-c81e5b). The break-even ladder remains the next step.
  - `LEAD-ICE-decomposition-probability-audit`: the s4_expr defect rerun (EV-SEMAEV-593b7a). It changed the ICINV m4 and FIB C16-1 outputs and was inert in every capped cell. No frozen conclusion changes.
  - `LEAD-ICP-isogeny-transported-relation-yield`: the CM-family small-x window base is exchangeable with a random base to ±0.04% at C24 (EV-CMFB-e3d1ec, toy, preliminary). The CM structure itself is untested.
  - Component improvements, all engineering, with rho ahead in each:
    - Joux–Vitse cover IC Hilbert stop, paired S ratio 1.85–1.88×, validated with caveats (`LEAD-TRF-jv-sieve-fp6-weak-class`);
    - char-2 D_4 Frobenius root finding, 0.854 and 0.815, an IC lower bound (`LEAD-ICB-composite-degree-c2pnb`);
    - aimed-subset gather, collect phase 1.093× established (`LEAD-ICB-koblitz-setup-phase-cost`);
    - k = 5 sum table at 0.22–0.70× the mitm incumbent on holdout curves, generic track, not admitted (`LEAD-ICP-generic-table-decomposition`).
  - Negative diagnostic: the Hecke D39 rank closure is reproduced by doubling rows alone (`LEAD-ICP-lattice-ic-relation-construction`).
  - Problem records and decisions attached to their leads: NISTSEED, PFDR theory rulings, BINSTD, FROB, ICEX, PEXT, the known-label index, the quartic-review repairs and Koblitz modular round 22.
- **Not about ECDLP:** PQC ideas and hypotheses (ML-KEM, ML-DSA, LWE, FN-DSA, SLH-DSA, HQC, NTRU), CI and runner repair reports, process problems, and session logs.

family	leads	closed	open, next step	blocked	unexamined
generic	26	10	11	2	3
ic-binary	22	11	10	1	0
ic-engine	19	5	8	5	1
ic-prime	26	12	6	5	3
transfer	23	14	5	2	2
structure	17	14	3	0	0
exotic	20	8	10	2	0
all	**153**	**74**	**53**	**17**	**9**

Files

- `dispositions.svg`, `dispositions.json`, `figures.py`: the figure, its counts, and the script that draws both from `knowledge/leads/ecdlp/`.
- `atlas-snapshot.md`: `python3 tools/lead_atlas.py --render` at this review.
- `report.pdf`, `build_pdf.py`: this README as a PDF, with the figure redrawn from `dispositions.json`.